higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING PRACTICE AND MAINTENANCE
NQF LEVEL 3

(6021003)

7 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Read ALL the questions carefully.
2. Answer ALL the questions.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Write neatly and legibly.

NCV

QUESTION 1: TECHNOLOGICAL SOLUTIONS IN ENGINEERING

1.1 One of the first steps in the engineering design process is to know what types or forms of information you can consult to research solutions.

Explain each of the following forms of written or printed information:

1.1.1 Manuals

1.1.2 Textbooks

1.1.3 Technical reports

1.1.4 Manufacturer specifications

(4 × 1) (4)

1.2 A certain fruit-packing company wants to improve its conveyer belt system for better transport of fruit.

Give FOUR design problems you will consider to solve the company's problem on the existing system. (4)

1.3 You are a design engineer in a truck building company. Tests must be done before designs can be put on the production line.

Name FOUR different tests and evaluations you will perform on the new design of a truck. (4)

1.4 A technician has discovered a problem with an existing design.

Write a report giving FOUR good reasons to justify why changes should be made to the existing product. (4)

1.5 Differentiate between the following:

1.5.1 Design for quality

1.5.2 Design for export

(2 × 1) (2)

1.6 Engineers search for improvement in their solutions in two ways, namely through modifications and by upgrading their design solutions.

Explain the terms *modifications* and *upgrades*. (2)

[20]

Copyright reserved

QUESTION 2: SYSTEMATIC FAULT-FINDING TECHNIQUES

2.1 Choose an item from COLUMN A that matches a description in COLUMN B. Write only the letter (A–H) next to the question number (2.1.1–2.1.8) in the ANSWER BOOK.

Vertical and Horizontal lines be 1½

COLUMN A		COLUMN B	
2.1.1	Fault	A	acceptable boundaries within which a system or a product can operate or deviate from the set operational standard and can still work safely and well
2.1.2	Operational test	B	a product or system that does not perform as expected
2.1.3	Systematic fault-finding techniques	C	dangers that can cause accidents, incidents and injuries
2.1.4	Failure	D	it is carried out to make sure that different components operate properly together and that the entire system performs as it should
2.1.5	Risks	E	when a part or component of the product or system has broken or is not working
2.1.6	Hazards	F	it is regulated and organised by step-by-step methods and processes that we use to find faults in a product, process or system
2.1.7	Tolerance level	G	directly looking and listening for faults
2.1.8	Direct observation	H	it can put people and the environment in danger of damage and injury

(8 × 1) (8)

2.2 A customer brings a car that has a faulty engine.

Which THREE methods would you use to identify the symptoms of the fault? (3)

Copyright reserved

2.3 Engineers use certain techniques to locate faults and determine their causes in a product or system.

State FIVE techniques to locate faults in the industry. (5)

2.4 Differentiate between the following:

2.4.1 A failure and a fault in a system

2.4.2 Relevant and irrelevant systems of a fault

(2 × 2) (4)

[20]

QUESTION 3: DIAGNOSIS OF MECHANICAL AND ELECTRICAL FAULTS IN ENGINEERING

3.1 Fault finding diagnostic involves a step-by-step and logical approach to find out which part or parts of a system is/are problematic.

Draw a flow diagram showing EIGHT principles that are used in fault-finding diagnostics. (8)

3.2 An engineer would have to use different tools or aids to locate and correct problems.

Name THREE commonly used tools or aids for locating a fault. (3)

3.3 You are employed as a field technician for a mechanical engineering company. A customer calls you to repair her faulty two-stroke petrol engine lawn mower.

List FIVE steps that you will use to correct the fault. (5)

3.4 Refer to QUESTION 3.3 above to prepare a fault finding diagnostic report for the management. (8 × ½) (4)

[20]

Copyright reserved

QUESTION 4: LUBRICANTS AND LUBRICATION SYSTEM

4.1 Differentiate between a *hydrostatic principle* and a *hydrodynamic principle*. (4)

4.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (4.2.1–4.2.8) in the ANSWER BOOK.

4.2.1 Natural lubricants consist of artificial chemicals that are artificially made and used instead of natural products (base oil).

4.2.2 A lubricant is any substance that is used to grease the surface of the different parts in a system that come into contact with each other in order to increase friction.

4.2.3 Viscosity is the degree to which a substance is thick and less able to flow.

4.2.4 Synthetic lubricants are mainly made of natural products such as vegetable, mineral or animal products.

4.2.5 The flash point is the lowest temperature at which a gas above a liquid will ignite when a flame is applied to it.

4.2.6 Dispersants are used to keep sludge, carbon and other substances from remaining in lubricants.

4.2.7 Fluid film occurs when two surfaces are completely separated by a coating of oil.

4.2.8 A clacking noise (like tapping on metal) in an engine may suggest that a piston is broken or damaged in some way.

(8 × 1) (8)

4.3 Describe the following types of lubrication systems as used in industry:

4.3.1 High-pressure lubricating system

4.3.2 Low-pressure lubricating system

4.3.3 Centrally distributed lubricating system

(3 × 2) (6)

4.4 Leaks can be a major source of wasted energy in a system. They can also lead to other operating losses.

Explain TWO such losses. (2)

[20]

Copyright reserved

QUESTION 5: MAINTENANCE OF PLANT AND EQUIPMENT IN ENGINEERING

5.1 An engineering plant and engineering equipment are governed by the policies, rules and principles of the company that owns them.

List FOUR principles on which an engineering plant and equipment operate. (4)

5.2 The Occupational Health and Safety Act (OHSA) 1993 (Act 85 of 1993) sets out general health and safety rules that must be followed by all employers and employees in different occupations.

Indicate FOUR rules that you need to know and apply in your workplace. (4)

5.3 Draw a flow diagram indicating FOUR safety measures to be followed when working with hazardous equipment. (4)

5.4 Engineering equipment may be dangerous if used incorrectly.

Give FOUR reasons for using equipment correctly. (4)

5.5 Differentiate between *engineering plant* and *engineering equipment*. (4)
[20]

TOTAL: 100

NCV

Copyright reserved